Lamola Boitumelo, 229246346
Maswanganyi Nsovo, 224788213
Assignment 2 Final
Due Date: 16 March 2026

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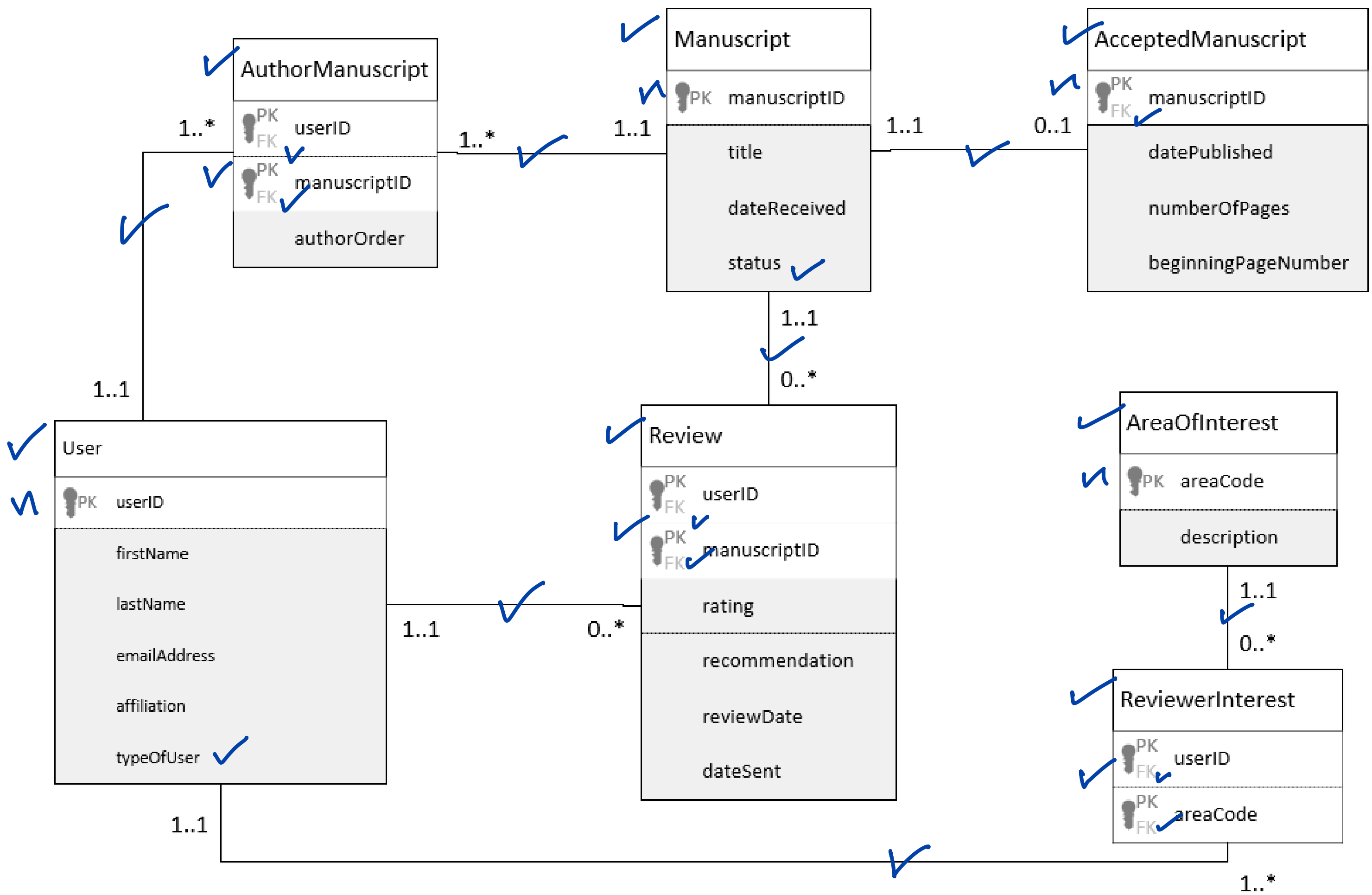
	Mark	Max
Entities	7	7
PK	5	5
FK	7	7
Subtype discriminators	2	2
Relationships	7	7
Attributes	2	2
	30	30

Modelling and Normalisation

1. Database design.

Updated version 1



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2.

[14] a) CarHire (carReg[pk][fk], cNo[pk][fk], make, cName, cPhone, hireDate, outletNo, outletLoc) 1

b) Desirable dependencies:

- carReg, cNo → make, cName, cPhone, hireDate, outletNo, outletLoc

Undesirable dependencies

Transitive dependency:

- outletNo → outletLoc

Partial dependency: 8

- carReg → make
- cNo → cName, cPhone

c) Normalisation

- CAR(carReg[pk], make)
- CUSTOMER(cNo[pk], cName, cPhone)
- CAR HIRED(carReg[pk][fk], cNo[pk][fk], hireDate,outletNo). 5
- OUTLET(outletNo[pk], outletLoc)

(7) 3. The dependency diagram is in the First Normal Form(1NF), since the NMU data provided is arranged in a **table format**, with **no repeating groups**, as well as an **established composite primary key** (PK) of barCode, empNo.

Dependencies are as follows:
Desired dependency: (barCode, empNo → itemDesc, rmNo, bldgNo, bldName, mgrName)

Undesired dependencies
Partial dependencies: - (barCode → itemDesc, rmNo)
- (empNo → mgrName)
Transitive dependency: (bldgNo → bldName)